

Diabetes Risk Prediction — ML Pipeline & REST API

End-to-end ML classification system: 86.7% accuracy, F1: 0.84, AUC-ROC: 0.91

[Tech Stack](#) | Python · Scikit-learn · Random Forest · Flask · REST API · SMOTE · Postman

Machine Learning

Flask API

Health-Tech

Classification

Production-Ready

Project Overview

- An end-to-end machine learning pipeline for diabetes risk classification, productionised as a Flask REST API ready for health-tech frontend integration. The pipeline features custom feature engineering, SMOTE oversampling to handle class imbalance, k-fold cross-validation, and a Random Forest classifier achieving 86.7% accuracy with F1: 0.84 and AUC-ROC: 0.91.

Key Technical Highlights

- Custom feature engineering pipeline: interaction terms, BMI-age buckets, and glucose-insulin ratio features derived from raw inputs.
- SMOTE (Synthetic Minority Oversampling Technique) applied to address class imbalance in the training set — preventing biased predictions.
- k-Fold cross-validation (k=10) used for robust model evaluation, ensuring accuracy metrics generalise beyond the training set.
- Random Forest classifier tuned via grid search over `n_estimators`, `max_depth`, and `min_samples_split` hyperparameters.
- Final model metrics: Accuracy 86.7%, F1-Score 0.84, AUC-ROC 0.91 — strong performance on an imbalanced medical dataset.
- Flask REST API with strict input validation, structured error responses, and a fully Postman-documented JSON schema.

Architecture & Design Decisions

- Data preprocessing pipeline: missing value imputation, outlier capping (IQR method), and MinMax scaling applied in sequence.
- SMOTE oversampling applied only on the training split after train/test split — prevents data leakage from test set.
- Scikit-learn Pipeline object chains preprocessing, SMOTE, and Random Forest estimator for reproducible, leak-proof inference.
- Flask API exposes a POST `/predict` endpoint accepting JSON patient features; returns risk probability and binary classification.

- Input validation layer rejects malformed requests with descriptive error messages before they reach the ML model.
- Model serialised with joblib for fast loading at API startup; Postman collection documents all request/response schemas.

Outcomes & Impact

- Achieved 86.7% accuracy and AUC-ROC of 0.91 — demonstrating strong generalisation on held-out test data.
- Production-ready REST API with documented JSON schema enables direct plug-in to any health-tech frontend or mobile app.
- Rigorous ML pipeline design (no data leakage, cross-validated metrics) makes results reliable and reproducible.
- Demonstrated full ML lifecycle ownership: data prep, feature engineering, model selection, evaluation, and API productionisation.